LIBRARY MANAGEMENT SYSTEM

OOPS PROJECT REPORT

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INTRODUCTION

A Library Management System is a software application used to manage library activities such as adding books, searching for books, issuing books, and recording returns. Maintaining these records manually can be time-consuming and may cause errors. This project provides a simple computerized solution to organize library information and improve efficiency.

The system uses Object-Oriented Programming (OOPS) concepts to divide the application into classes and objects. It helps maintain book records, student details, and transaction information in an organized manner.

PROBLEM STATEMENT

Manual library management makes it difficult to search for books, check availability, maintain student records, and track issues and return dates. It can also lead to duplicate entries and misplaced records. The proposed system reduces these problems by maintaining organized digital records.

OBJECTIVES

- Maintain book and student records.
- Search for books by ID, title, or author.
- Issue books for registered students.
- Record returned books.
- Track book availability.
- Reduce paperwork and manual errors.
- Apply OOPS concepts to a real-world application.

REQUIREMENTS

Hardware: Computer, minimum 4 GB RAM, keyboard, mouse, and monitor.

Software: Windows/macOS/Linux, C++ or Java, a suitable IDE, and file handling or database storage.

SYSTEM DESIGN

The system contains the following classes:

- **Book:** Stores book ID, title, author, and availability. Functions include adding, displaying, updating, and searching for book details.
- **Student:** Stores student ID, name, course, and contact details. It manages member information.
- **Librarian:** Manages book records, registers students, and performs issues and return operations.
- **Library:** Organizes books and members and coordinates library activities.
- **Transaction:** Stores transaction ID, book ID, student ID, issue date, due date, and return date.

Each class performs a specific task, making the program more modular and easier to maintain.

OOPS CONCEPTS USED

Classes and Objects: A class is a blueprint, while an object is its instance. For example, Book is a class, and each individual book can be an object.

Encapsulation: Combines data and functions in one class and controls access to data using suitable access modifiers.

Inheritance: Allows one class to reuse the properties and functions of another class. For example, Student and Librarian can inherit common details from a Person class.

Polymorphism: Allows the same function interface to perform different tasks. For example, display Details () can display books or student information.

Abstraction: Hides internal implementation details and shows only essential operations to the user.

Constructors: Initialize objects with starting values when they are created.

These concepts improve code reusability, organization, and maintainability.

MAIN MODULES AND WORKING

Book Management: Adds, updates, and displays book records.

Search Module: Finds books using their ID, title, or author.

Student Management: Maintains registered member information.

Book Issue: Checks book availability and member details, records the issue and due dates, and updates the book status.

Book Return: Records the return date and makes the book available again.

Record Management: Stores and organizes book, student, and transaction information using the selected storage method.

System Workflow

1. The librarian starts the application.
2. The system displays available operations.
3. The librarian selects an operation.
4. The system processes book or student information.
5. For issuing a book, availability is checked.
6. The transaction is recorded and book status is updated.
7. When a book is returned, the return record and availability are updated.

TESTING

Test Case	Expected Result
Add a book	Book record is saved
Search for a book	Matching details are displayed
Invalid book ID	Error message appears
Issue an available book	Transaction is recorded
Issue an unavailable book	Issue is rejected
Return a book	Status changes to available

These are expected results and should be verified against the completed program.

ADVANTAGES AND LIMITATIONS

Advantages:

- Saves time and reduces paperwork.
- Makes book searching easier.
- Organizes records systematically.
- Improves tracking of issued books.
- Demonstrates practical OOPS concepts.

Limitations:

- A basic version may not support online access.
- Automatic fine calculations may be unavailable.
- Barcode scanning and notifications may not be included.
- Security depends on the implemented access controls.

FUTURE SCOPE

The system can be improved by adding MySQL database integration, secure login, automatic fine calculation, online book reservations, barcode scanning, due-date reminders, and report generation.

CONCLUSION

The Library Management System is a practical application for Object-Oriented Programming. It simplifies book management, student record maintenance, and issue and returns operations. The

project demonstrates classes, objects, encapsulation, inheritance, polymorphism, and abstraction. It also improves understanding of software design and modular programming. The system can be expanded in the future to support the requirements of larger libraries.

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